

Programming Using C++ Level 1

Variable scope
Vs
Global variable

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Definition

- ◆ Variable scope is the part of the program where a variable can be accessed.
- ◆ A local variable is declared inside a function or block and can only be accessed within its scope.
- ◆ A global variable is declared outside all functions and can be accessed from different parts of the program.

Explanation

◆ This program has three variables with the same name `x`, but they belong to different scopes.

- Global `x` → declared outside all functions and has the value 30.

- Local `x` inside `main()` → has the value 10.

- Local `x` inside `MyFunction1()` → has the value 20.

◆ The local variables hide the global variable when they have the same name inside their own scope.

☰ Steps in main()

1☐ Declare the global variable

◆ Define x outside all functions with the value 30.

2☐ Declare a local variable in MyFunction1()

◆ Define another x inside MyFunction1() with the value 20.

3☐ Declare a local variable in main()

◆ Define another x inside main() with the value 10.

4☐ Access the local variable in main()

◆ cout uses the x declared inside main().

5☐ Call MyFunction1()

◆ The function uses its own local x.

6☐ Access the global variable

◆ Use ::x to explicitly access the global x.

💡 Example

◆ Global variable:

```
int x = 30;
```

◆ Local variable:

```
int x = 10;
```

◆ Accessing the global variable when a local variable has the same name:

➡ Result

◆ The program displays:

The value of x inside main is: 10

The value of x inside function is: 20

The Global value of x is: 30

Note

- ◆ When a local variable has the same name as a global variable, the local variable takes priority within its scope.
- ◆ The scope resolution operator `::` can be used to access the global variable.
- ◆ In this example, `::x` refers specifically to the global `x`.

⊘ Warning

⚡ Using the same variable name for local and global variables can make code harder to understand.



Comparison

Local Variable

- ✓ Declared inside a function or block.
- ✓ Can only be accessed within its scope.
- ✓ Usually exists while its function or block is executing.

Global Variable

- ✓ Declared outside all functions.
- ✓ Can be accessed from different functions.
- ✓ Exists throughout the execution of the program.

✦✦ Summary

- ✓ Local variables are limited to their own scope.
- ✓ Global variables can be accessed from different parts of the program.
- ✓ The `::` operator can be used to access a global variable when a local variable has the same name.

Thank you 😊